

8.2 Diffraction & Interference

Question Paper

Course	CIE A Level Physics
Section	8. Superposition
Topic	8.2 Diffraction & Interference
Difficulty	Medium

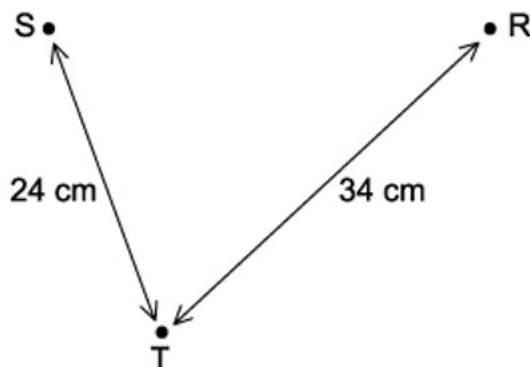
Time allowed: 20

Score: /10

Percentage: /100

Question 1

Two wave generators are placed at points S and R to produce water waves with an identical wavelength. At point T both waves have the same amplitude. The distances from S to T and R to T are shown on the diagram below



As the student walks between S to T she notices that the loudness of the sound increases then decreases repeatedly.

What is the wavelength of the waves when the wave generators are in phase and the amplitude at T is zero?

- A. 6 cm
- B. 4 cm
- C. 3 cm
- D. 2 cm

[1 mark]

Question 2

A diffraction grating with 1000 lines per cm has light passed through it. An identical light source is also passed through two slits 0.5 mm apart.

Both will produce intensity maxima and minima on a screen.

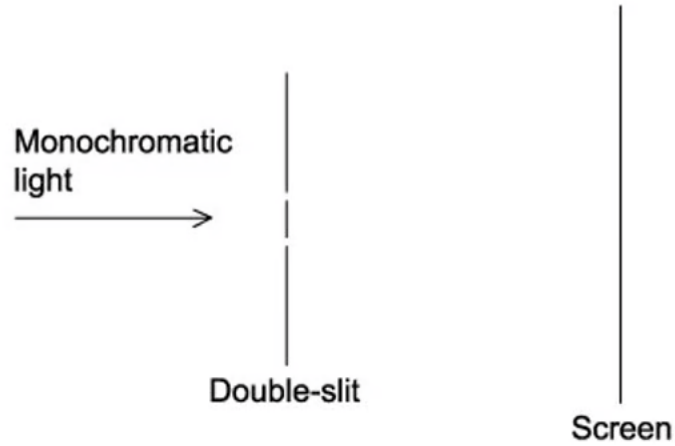
Which of the following statements about the separation and sharpness of the maxima is correct?

- A. the diffraction grating maxima are more widely spaced and are less sharp than the two-slit maxima
- B. the diffraction grating maxima are less widely spaced and are sharper than the two-slit maxima
- C. the diffraction grating maxima are more widely spaced and are sharper than the two-slit maxima
- D. the diffraction grating maxima are less widely spaced and are less sharp than the two-slit maxima

[1 mark]

Question 3

The effect of a double slit on monochromatic light is observed by set up in the diagram



Interference fringes are seen on the screen.

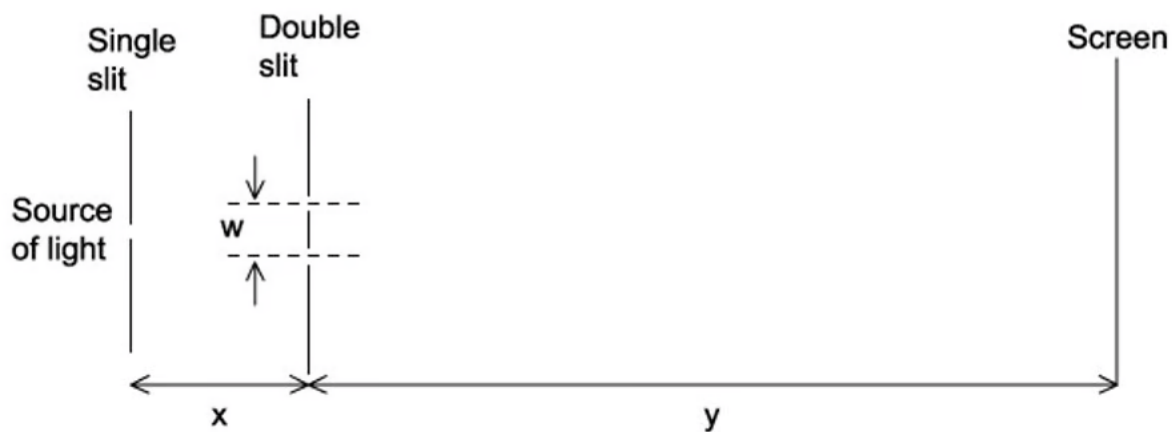
Which of the following changes would increase the distance between the adjacent fringes?

- A. decrease the width of each slit
- B. decrease the distance between the two slits
- C. use light of a higher frequency
- D. move the screen closer to the double-slit

[1 mark]

Question 4

The equipment was set up as in the diagram to investigate the interference of a double slit.



Which of the following changes would increase the fringe spacing?

- A. decreasing the wavelength of the light
- B. decreasing distance y
- C. decreasing distance x
- D. decreasing the distance w

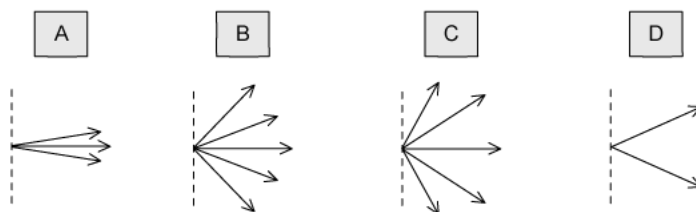
[1 mark]

Question 5

A beam of monochromatic light is directed at a diffraction grating as shown in the diagram



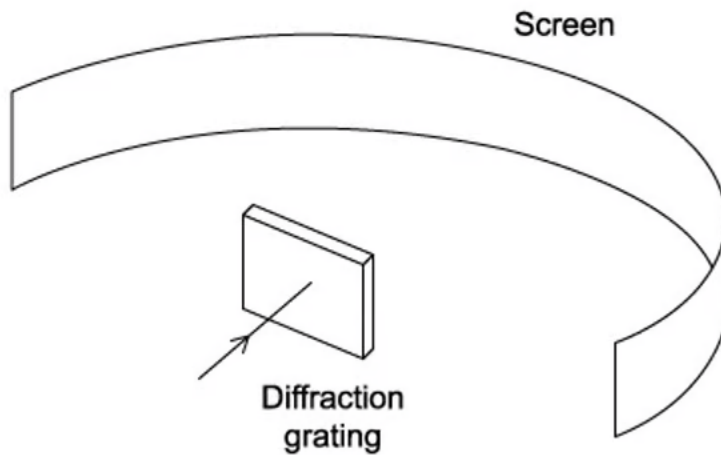
Which of the following diagrams show the possible directions of the light that will give maximum intensity?



[1 mark]

Question 6

A beam of monochromatic light with a wavelength of 690 nm is directed at a diffraction grating with 300 lines per mm as shown in the diagram



This set up produces a series of maxima on the screen, what is the greatest number that can be observed?

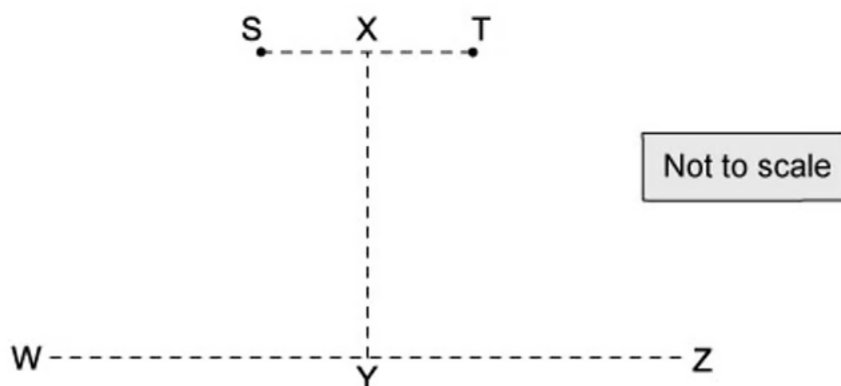
- A. 9
- B. 8
- C. 5
- D. 4

[1 mark]

Question 7

A student is investigating the interference of waves. They set up coherent waves produced at points S and T and travel outwards in all directions.

The line XY is halfway between S and T and perpendicular to the joining line S and T. The distance XY is much greater than the distance ST.



Which line or lines would an interference pattern be seen?

- A. WZ only
- B. XY only
- C. Both XY and WZ
- D. Neither XY or WZ

[1 mark]

Question 8

Certain circumstances will produce visible interference fringes from two light sources.

Which of the following conditions will enable visible interference fringes to be seen?

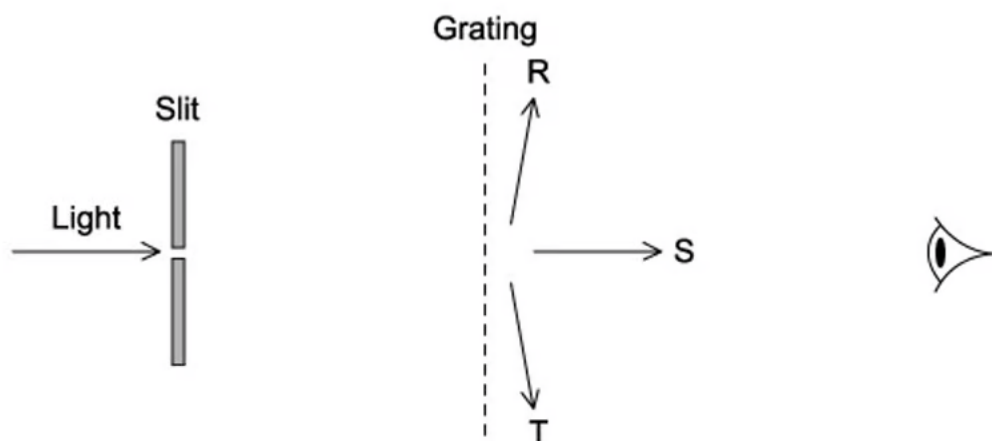
- A. using one light which is polarised at right angles to light from the other source
- B. using incoherent sources
- C. using a blue light source
- D. using sources from which the light does not overlap

[1 mark]

Question 9

Monochromatic light with a wavelength of 600 nm is observed through a diffraction grating with 500 lines per mm.

The light is passed through a narrow slit the lines from the slit is parallel to the grating. The light goes through the slit then the grating.



The observer changes position to; R – parallel with the grating, S – opposite the slit, T – parallel to the grating on the opposite side.

What is the number of images he would see?

- A. 7
- B. 6
- C. 4
- D. 3

[1 mark]

Question 10

Monochromatic light produces interference fringes with a separation of x on a screen with distance D with a pair of slits with separation a .

The student doubles a and D .

What would be the new fringe separation?

A. $2x$

B. $4x$

C. x

D. $\frac{x}{2}$

[1 mark]

Model Answers: Medium

1

The correct answer is **B** because:

- Path difference is defined as the difference between the lengths of two paths of the two different waves
 - So, the path difference is equal to $RT - ST = 34 - 24 = 10 \text{ cm}$
- When two sources of waves that are in phase interfere:
 - At maxima: the path difference is equal to $n\lambda$, constructive interference will be produced at T
 - At minima: the path difference is $\left(n + \frac{1}{2}\right)\lambda$, destructive interference will be produced at T
- At T, the amplitude is zero, so, a minima is produced, indicating that destructive interference occurs
 - So, path difference $= \left(n + \frac{1}{2}\right)\lambda$
- We're told the loudness of the sound increases then decreases repeatedly, so we can insert values of n (larger than zero) until we find a whole number of wavelengths
 - If $n = 0$, wavelength $\lambda = \frac{10}{\left(0 + \frac{1}{2}\right)} = 20 \text{ cm}$
 - If $n = 1$, wavelength $\lambda = \frac{10}{\left(1 + \frac{1}{2}\right)} = 6.67 \text{ cm}$
 - If $n = 2$, wavelength $\lambda = \frac{10}{\left(2 + \frac{1}{2}\right)} = 4 \text{ cm}$
 - If $n = 3$, wavelength $\lambda = \frac{10}{\left(3 + \frac{1}{2}\right)} = 2.86 \text{ cm}$
- Therefore, when $n = 2$, the wavelength is 4 cm

2

The correct answer is **C** because:

- The spacing between two lines in the diffraction grating is equal to $\frac{1}{1000} \text{ cm} = 0.01 \text{ mm}$
- The separation of the 2 narrow slits is equal to $= 0.5 \text{ mm}$
 - So, the diffraction grating maxima are more widely spaced as the spacing between the lines in the diffraction grating is less than the separation of the 2 narrow slits
- Consider the spots formed to be black and white: a higher sharpness would cause the spots formed to appear 'more white' or 'more black'
 - This means they can be more distinguished from each other
- Additionally, the diffraction grating maxima are sharper than the two-slit maxima since more interference occurs from the waves that get diffracted from more slits in the diffraction grating
- Hence, the amplitudes of more diffracted waves contribute to every single maxima
- Therefore, the diffraction grating maxima are more widely spaced and are sharper than the two-slit maxima

A is incorrect because two-slit interference results in sharper maxima since more waves are diffracted which contribute to each maxima

B & D is incorrect because the diffraction grating maxima are more widely spaced as the spacing between the lines in the diffraction grating is less than the separation of the 2 narrow slits

3

The correct answer is **B** because:

- The fringe separation equation is given by
 - $x = \frac{\lambda D}{a}$
- Consider option B:
 - If we decrease the distance a between the two slits, this will increase the fringe separation x

A is incorrect because the width of each slit is not a quantity in the fringe spacing equation

C is incorrect because using light of a higher frequency will result in a smaller wavelength λ , and hence cause a decrease in fringe spacing x

D is incorrect because moving the screen closer to the double-slit will decrease the value of D and hence the value of fringe spacing x will decrease

4

The correct answer is **D** because:

- The fringe separation equation is given by
 - Fringe separation = $\frac{\lambda y}{w}$
- Consider option D:
 - Decreasing the distance w , or slit separation, causes the fringe separation to increase

A is incorrect because decreasing the wavelength λ of the light will cause the fringe separation to decrease

B is incorrect because decreasing the distance y between the slits and the screen will cause the fringe separation to decrease

C is incorrect because the distance x is not a quantity in the fringe spacing equation

5

The correct answer is **B** because:

- The maximum intensity of light is the region where the light is brightest due to constructive interference
- The region of maximum intensity is always in the direction in which the light is incident
- Additionally, maximum intensities are not limited to the region just mentioned and regions close to it, but maximum intensity can also be observed at different angles

- There are different orders which are given by the diffraction grating equation is given by : $d \sin \theta = n\lambda$
- So, the angles of diffraction of the maxima are equal to $\theta_n = \sin^{-1}\left(\frac{n\lambda}{d}\right)$
 - When $n = 0$, $\theta_0 = \sin^{-1}(0) = 0$
 - When $n = 1$, $\theta_1 = \sin^{-1}\left(\frac{\lambda}{d}\right)$
 - When $n = 2$, $\theta_2 = \sin^{-1}\left(\frac{2\lambda}{d}\right)$
 - When $n = 3$, $\theta_3 = \sin^{-1}\left(\frac{3\lambda}{d}\right)$
- The angle of diffraction from the 0th order maxima is smaller than the angular separation (the difference in angle of diffraction) of the 2nd order and 1st order maxima, i.e.
 - $\theta_1 - \theta_0 = \theta_1 < (\theta_2 - \theta_1)$
- The same is true as the order n increases, so, for small angles, the difference in angles would be the same

A is incorrect because regions of maximum intensity should be observed at different angles, the range of angles in this diagram is too small

C is incorrect because the angles of diffraction in this diagram are too large

D is incorrect because the region of maximum intensity is always in the direction in which the light is incident, meaning there must be a central arrow, which this diagram does not have

6

The correct answer is **A** because:

- The diffraction grating equation is given by
 - $d \sin \theta = n\lambda$
- Where:
 - Slit separation $d = \frac{1}{300} = 3.33 \times 10^{-3} \text{ mm} = 3.33 \times 10^{-6} \text{ m}$
 - Wavelength $\lambda = 690 \times 10^{-9} \text{ m}$
 - Number of bright fringes on one side = n
- The maximum angle θ from the central fringe is 90° on each side (it is actually less

- but we cannot have a value of θ being greater than 90°)
- Rearranging for n :
 - $d \sin \theta = n\lambda$



$$\sin \theta = \frac{a}{\lambda}$$

- When $n=3$, $\theta_3 = \sin^{-1}\left(\frac{3\lambda}{d}\right)$
- The angle of diffraction from the 0th order maxima is smaller than the angular separation (the difference in angle of diffraction) of the 2nd order and 1st order maxima, i.e.
 - $\theta_1 - \theta_0 = \theta_1 < (\theta_2 - \theta_1)$
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 - Wavelength $\lambda = 690 \times 10^{-9} \text{ m}$
 - Number of bright fringes on one side = n
- The maximum angle θ from the central fringe is 90° on each side (it is actually less — but we cannot have a value of θ being greater than 90°)
- Rearranging for n :
 - $d \sin \theta = n\lambda$
 - $n = \frac{d \sin \theta}{\lambda} = \frac{(3.33 \times 10^{-6}) \sin 90^\circ}{690 \times 10^{-9}} = 4.8$
- This means, the 4th bright fringe can be obtained but not the 5th bright fringe since we cannot have 4.8 bright fringes, so, we can have up to 4 bright fringes on one side

- We will also have 4 bright fringes on the other side and one central fringe, these are also maxima
 - Therefore, the total number of fringes = $4 + 1 + 4 = 9$

7

The correct answer is **A** because:

- An interference pattern consists of a series of alternating constructive interference bright spots and destructive interference dark spots
 - An interference pattern will not be observed along a line where only constructive or destructive interference occurs
- The waves are produced at S and T – the phase difference between the waves from S and T is constant along line XY since the waves are coherent and XY is midway from S and T
- Therefore, the path difference along XY is always zero – that means, only constructive interference occurs along this line (for constructive interference, path difference where $n = 0, 1, 2, 3$ etc.)
- So, there can be no interference pattern along XY
 - Therefore, an interference pattern will only be observed along line WZ

8

The correct answer is **C** because:

- In order to observe interference in light waves, the following conditions must be met:
 - The source must be coherent – they must have the same phase
 - The source must be monochromatic – the light must be of a single wavelength
- Using blue light of a single wavelength satisfies the second condition

A is incorrect because light will only interfere with a light of the same polarisation

B is incorrect because light will only interfere if both sources are coherent

D is incorrect because the light sources must overlap in order for interference to occur

9

The correct answer is **A** because:

- The diffraction grating equation is given by
 - $d \sin \theta = n\lambda$
- Where:
 - Slit separation $d = \frac{1}{500} = 2.0 \times 10^{-3} \text{ mm} = 2.0 \times 10^{-6} \text{ m}$
 - Wavelength $\lambda = 600 \times 10^{-9} \text{ m}$
 - Number of images on one side = n
- The maximum angle θ from the central fringe is 90° on each side (it is actually less – but we cannot have a value of θ being greater than 90°)
- Rearranging for n :
 - $d \sin \theta = n\lambda$
 - $n = \frac{d \sin \theta}{\lambda} = \frac{(2.0 \times 10^{-6}) \sin 90^\circ}{600 \times 10^{-9}} = 3.3$
- This means, the 3rd image can be seen but not the 4th image since we cannot have 3.3 images, so, we can have up to 3 images on one side
- We will also have 3 bright fringes on the other side and one central fringe, these are also maxima
 - Therefore, the total number of fringes = $3 + 1 + 3 = 7$

10

The correct answer is **C** because:

- The fringe separation equation is given by
 - $x = \frac{\lambda D}{a}$
- Where:
 - x is the fringe separation
 - λ is the wavelength
 - $2D$ is the distance between the slits and the screen
 - $2a$ is the slit separation
- So, the new fringe spacing is equal to:
 - $x = \frac{\lambda(2D)}{(2a)} = \frac{\lambda D}{a}$
- Therefore, the value of the fringe spacing x does not change

